

Def. Let F be a field. Define the General Linear Group as:

$$GL_n(F) \stackrel{\text{def}}{=} \{A \in M_{n \times n}(F) \mid \det A \neq 0\}$$

$I_n \in GL_n(F)$, $\det A, \det B \neq 0 \Rightarrow \det AB \neq 0$, $\det A \neq 0 \Rightarrow \det A^{-1} \neq 0$, exists

Thus it is group.

Def. Define the Special Linear Group as: $SL_n(F) \stackrel{\text{def}}{=} \{A \in M_{n \times n}(F) \mid \det A = 1\}$

Lemma. Let $\mathbb{F}_q$ be a finite field of order q . Then,
 $GL_n(\mathbb{F}_q)$ has order $(q^n - 1)(q^n - q)(q^n - q^2) \cdots (q^n - q^{n-1})$

$SL_n(\mathbb{F}_q)$ has order $\frac{1}{q-1} |GL_n(\mathbb{F}_q)|$

Proof. Clearly, $|M_{n \times n}(\mathbb{F}_q)| = q^{n^2}$.

Choose the first column as non-zero, there are $q^n - 1$ possibilities.

And, the second column all but a multiple of the first, then $q^n - q$ possibilities.

Successively, choose the third as all but not linear combination of first and second, then there are $q^n - q^2$ (More specifically, q^2 is the number of cases of $a_1u_1 + a_2u_2$).

Inductively, we obtain that $|GL_n(\mathbb{F}_q)| = \prod_{k=0}^{n-1} (q^n - q^k)$

Meanwhile, observe that

$$\varphi: GL_n(\mathbb{F}_q) \rightarrow \mathbb{F}_q^\times; A \mapsto \det A$$

is well-defined (because $\det A$ is obtained by operations on $\mathbb{F}_q$), onto ($\det \begin{pmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{pmatrix} = 1$) and Homomorphism ($\det AB = \det A \cdot \det B$) with $\ker \varphi = SL_n(\mathbb{F}_q)$. Since

$$GL_n(\mathbb{F}_q) / \ker \varphi \cong \mathbb{F}_q^\times, \text{ we obtain } |\ker \varphi| = |SL_n(\mathbb{F}_q)| = \frac{1}{q-1} \cdot |GL_n(\mathbb{F}_q)|.$$

$$(p-1)^2(p+1).$$

Problem.

- 1). Find order of $GL_2(\mathbb{F}_p)$, where p is the prime number.
- 2). Determine $P = \left\{ \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix} \mid a \in \mathbb{F}_p \right\}$ is Sylow p -subgroup of $GL_2(\mathbb{F}_p)$.
- 3). Find order of $N_{GL_2(\mathbb{F}_p)}(P)$ and $\# \text{Syl}_p(GL_2(\mathbb{F}_p))$.

Proof. $A \in GL_2(\mathbb{F}_p) \Leftrightarrow \det A \neq 0$.

The possible vector in the first row in A is $p^2 - 1$, except zero vector.
number of

For chosen first row, the possible number of second row is $p^2 - p$,
except scalar multiplication of the first row so,

$$|GL_2(\mathbb{F}_p)| = (p^2 - 1)(p^2 - p) = p \cdot (p-1)^2(p+1) = p(p^3 - p^2 - p + 1)$$

Clearly P has order of p , and $p \nmid p^3 - p^2 - p + 1$, so $P \in \text{Syl}_p(G)$.

Find $N_G(P)$. Since for any $\begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \in GL_2$, where $ac \neq 0$,

$$\begin{aligned} & \begin{pmatrix} c & -b \\ 0 & a \end{pmatrix} \cdot \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \cdot \frac{1}{ac} \\ &= \begin{pmatrix} c & -b \\ 0 & a \end{pmatrix} \begin{pmatrix} a & b + ck \\ 0 & c \end{pmatrix} \cdot \frac{1}{ac} \\ &= \begin{pmatrix} ac & cb + c^2k - bc \\ 0 & ac \end{pmatrix} \cdot \frac{1}{ac} \\ &= \begin{pmatrix} 1 & cka^{-1} \\ 0 & 1 \end{pmatrix} \in P, \text{ therefore } |N_G(P)| = (p-1)^2 \cdot p. \end{aligned}$$

$$\text{So, } n_p = [GL_2 : N_{GL_2}(P)] = p+1. \quad \square$$

Def. The Heisenberg group over F defined by

$$H(F) \stackrel{\text{def}}{=} \left\{ \begin{pmatrix} 1 & a_1 & a_2 \\ 0 & 1 & a_3 \\ 0 & 0 & 1 \end{pmatrix} \mid a_i \in F \right\} \leq SL_3(F).$$

Prop. $Z(H(F)) \cong (F, +)$

Proof observe that

$$\underbrace{\begin{pmatrix} 1 & a_1 & a_2 \\ 0 & 1 & a_3 \\ 0 & 0 & 1 \end{pmatrix}}_A \cdot \underbrace{\begin{pmatrix} 1 & b_1 & b_2 \\ 0 & 1 & b_3 \\ 0 & 0 & 1 \end{pmatrix}}_B = \begin{pmatrix} 1 & a_1+b_1 & a_2+b_2+a_1b_3 \\ 0 & 1 & a_3+b_3 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\text{Therefore, } \begin{pmatrix} 1 & a_1 & a_2 \\ 0 & 1 & a_3 \\ 0 & 0 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & -a_1 & -(a_2 - a_1 a_3) \\ 0 & 1 & -a_3 \\ 0 & 0 & 1 \end{pmatrix}$$

And, $AB = BA \Leftrightarrow a_1 b_3 = a_3 b_1$. This implies

$$A \in Z(H(F)) \Leftrightarrow \forall b_1, b_3 \in F, a_1 b_3 = a_3 b_1 \\ \Leftrightarrow a_1 = a_3 = 0.$$

$$\text{Therefore, } Z(H(F)) = \left\{ \begin{pmatrix} 1 & 0 & a \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \mid a \in F \right\}.$$

Now, $F \rightarrow Z(H(F)) : a \mapsto \begin{pmatrix} 1 & 0 & a \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ is isomorphism, by

$$\begin{pmatrix} 1 & 0 & a \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & b \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & a+b \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

$$GL_3(\mathbb{F}_q) \quad (q^3-1)(q^3-q)(q^3-q^2)$$

$$| \quad | \\ SL_3(\mathbb{F}_q) \quad (q^3-1)(q^3-q)q^2$$

$$| \quad | \\ H(\mathbb{F}_q) \quad q^3$$

$$| \quad | \\ Z(H(\mathbb{F}_q)) \cong (\mathbb{F}_q, +) \quad q$$

$$| \quad | \\ 1.$$